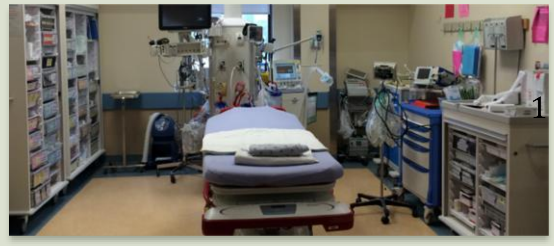
Ruptured AAA



Section I: Scenario Demographics

Scenario Title:	Ruptured Abdominal Aortic Aneurysm
Date of Development:	13/12/2014 (DD/MM/YYYY)
Target Learning Group:	☒ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
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Section III: Curriculum Integration

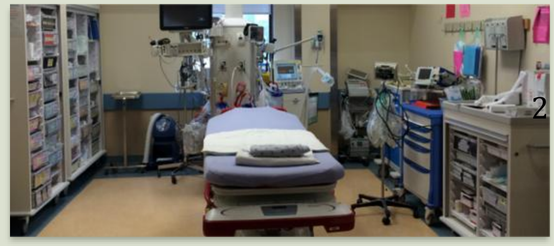
Learning Goals & Objectives	
Educational Goal:	To simultaneously diagnose and treat a patient with undifferentiated shock.
CRM Objectives:	1) Effectively establish oneself as the team leader and exercise non-verbal leadership techniques 2) Prepare and expedite the transfer of a critical patient from the ED to the operating room
Medical Objectives:	1) Explore the use of ultrasound in a patient with undifferentiated shock. 2) Appreciate the controversy of "normalization of vital signs" as a target in the resuscitation of a ruptured AAA through the utilization of permissive hypotension.

Case Summary: Brief Summary of Case Progression and Major Events

A 70-year-old male presents to the emergency department after a syncopal episode and then being found obtunded by his daughter. He is hypotensive and tachycardic on arrival secondary to a AAA rupture into the retroperitoneal space. He requires intubation and fluid resuscitation with blood products to avoid a PEA arrest secondary to hypovolemia.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case
You are working an evening shift at a tertiary care emergency department. You receive a call from a paramedic to alert you to the arrival of a 70-year old male who had a syncopal episode and was then found to be obtunded by his daughter. The patient is now in the resuscitation bay.

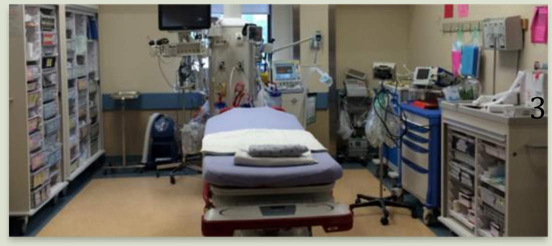
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most*	☐ Emotional/Experiential
	☐ Hybrid	*important*	☐ Other:
	☐ Task Trainer	*dimension(s)*	☐ N/A
Confederates	**Brief Description of Role**		
Daughter	Gives past medical history, medication history and review of systems.		

C. Required Monitors		
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☒ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage
Right flank bruising.

F. Approximate Timing					
Set-Up:	5 min	Scenario:	12 min	Debriefing:	15 min



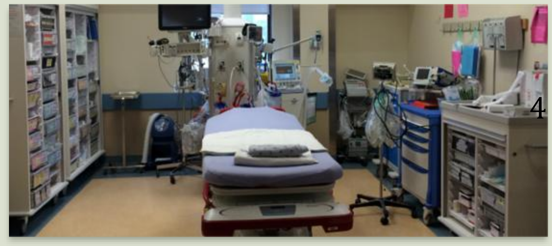
Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Jacques Patrice	Age: 70		Weight: 100kg
Gender: ☒ M ☐ F	Code Status: Full Code		
Chief Complaint: Found obtunded.			
History of Presenting Illness: Daughter was visiting the patient his apartment; the patient had an episode of sudden loss of consciousness and postural tone and was then found to be obtunded and minimally responsive. He was breathing and eyes were intermittently opening to the daughter shaking his shoulders. Last seen well yesterday.			
Past Medical History:	HTN	Medications:	Lisinopril 20mg PO Daily
	Hyperlipidemia		Atorvastatin 20mg PO QHS
	T2DM		Metformin 500mg PO BID
Allergies: NKDA			
Social History: Smoker, 40 PY history			
Family History: Non-contributory			

Review of Systems: (from daughter)				
	CNS:	Obtunded, responded to painful stimuli		
	HEENT:	Normal		
	CVS:	Normal		
	RESP:	No cough		
	GI:	No diarrhea		
	GU:	No frequency, urgency, dysuria		
	MSK:	Normal	INT:	Noticed bruising on flank

B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 130/min	BP: 70/46	RR: 20/min	O_2SAT: 95%%
Rhythm: Sinus	T: 36.7°C	Glucose: 9.0 mmol/L	GCS: 8 (E 2 V 2 M4)
General Status: Obtunded, decreased mental status			

CNS:	Pupils 2+ reactive bilaterally, moving all 4 limbs to pain, no FND, normal reflexes		
HEENT:	Normal		
CVS:	Tachycardia, no EHS, weak pulse in all 4 extremities		
RESP:	Secretions pooling in oropharynx, GAEB		
ABDO:	Soft, non-tender, *bruising on right flank,* (if asked for, “palpable mass in mid abdomen”)		
GU:	Normal		
MSK:	Normal	SKIN:	Cool, clammy extremities



Section VI: Scenario Progression

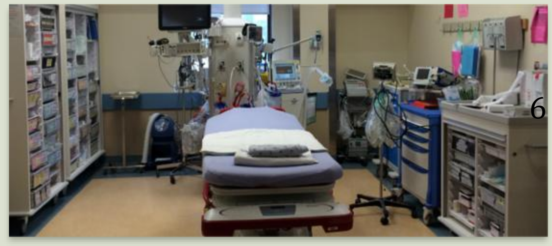
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 130/min BP: 70/46 RR: 12/min O_2SAT: 95% T: 36.7°C	Obtunded, GCS: 8 (E2V2M4)	Learner Actions -☐Monitor/Full vitals -☐Hx/PE -☐2 Large bore IVs -☐IV fluid bolus x 1L -☐Labs -☐Type and crossmatch -☐EKG -☐CXR -☐Bedside ultrasound	Modifiers *Changes to patient condition based on learner action* - IV Fluid Bolus x 1L→ HR to 125, BP to 80/50 Triggers *For progression to next state* - 5 minutes → **2. Deterioration**
2. Deterioration → BP: 60/P	Obtunded, GCS: 6 (E1V1M4)	Learner Actions -☐IV fluid bolus x 2 -☐Intubation -☐Peripheral vasopressors -☐PRBC infusion -☐MTP -☐Central Line insertion	Modifiers Triggers -If intubation occurs before 2nd bolus of NS or blood products → **3. PEA** - Initiation of blood products → **4. Resolution**
3. PEA → BP: 0/0 → O_2SAT: 0 (Rate and rhythm continue)	Unresponsive, No pulses felt	Learner Actions -☐ACLS PEA protocol -☐Timely recognition of PEA -☐CPR with minimal interruptions -☐Epinephrine 1mg IV -☐PRBC infusion -☐Crash Intubation	Modifiers - Intubation → ETCO$_2$, normal waveform and value of 12 Triggers - ≥2U PRBC or 2 rounds of CPR/Epinephrine → **4. Resolution**
4. Resolution → HR: 120/min → BP: 94/61 → RR: 12 or set by vent → O_2SAT: 95%	Obtunded, GCS:4 if not intubated. If intubated, intubated and sedated	Learner Actions -☐ Consultation (Vascular surgery/ICU) -☐Arrange to transfer to OR -☐Arrange advanced imaging (CTA) -☐MTP	**END SCENARIO PRN.**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 140	K: 4.0	Cl: 110	HCO_3: 16	BUN: 14	Cr: 185	Glu: 7.5
Ca:	Mg:	PO_4:	Albumin:			
VBG	pH: 7.13	PCO_2: 50	PO_2: 75	HCO_3: 16	Lactate:4.5	
WBC: 12	Hg: 85	Hct: .27	Plt: 122			

Images (ECGs, CXRs, etc.)	
ECG ECG source: http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg	Pre-intubation Xray Source: http://radiopaedia.org/images/220869
Post Intubation X-ray CXR source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg	

Ultrasound Video Files	
Normal Lung Sliding	Collapsible IVC
Hyperdynamic LV, no pericardial effusion	AAA+, no intraabdominal fluid (bleeding into retroperitoneum)



Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing
- What is the sequence that needs to occur to transfer a patient from our ED to the operating room? What needs to happen? How can we make this as efficient as possible? - What non-verbal techniques to good leaders exhibit? - What guides have been described regarding the use of ultrasound in undifferentiated shock? - Why was the bedside ultrasound negative for intra-abdominal fluid? - Describe the controversy of BP management in ruptured AAA?

Key Moments
- Ultrasound confirmation of a AAA
- PEA after intubation due to inadequate fluid resuscitation
- Transfer patient to OR